APPLIED R FOR PUBLIC HEALTH

Intro to R — Epidemiology & Statistics Resources

A curated reading and tooling list for doing epidemiology and biostatistics in R.

LEARNING OBJECTIVES

- Point the team to where the epidemiology and statistics behind the modelling live
- Favor **applied R for epidemiology** first, since this group already models and is learning R
- Curate vetted R packages for epi and biostatistics work
- Offer refreshers, references, and guidance on when to involve a statistician

CONTENTS

How to Use This Document

- The Day 4 lessons stayed on the **R mechanics** — how to fit a model and get its results out — and deliberately left the **statistics** (which model, what the estimate means) aside
- This document is the companion to that choice: a starting-point reading and tooling list for the epidemiology and biostatistics themselves

- It is organized so the most directly useful material for an R-learning epi team comes first; later sections are refreshers and references
- These are pointers, not endorsements of any single method — **method choice and interpretation should involve a statistician**

> **NOTE**
>
> The audience are public-health professionals who already practice epidemiology and already model. The emphasis here is therefore on **doing that work in R**, with foundational material included as references rather than as instruction.

DEEP DIVE

Nothing in this list is required to follow the course; it exists so the team can go deeper on their own and so the modelling day could stay light on statistics. Treat the first section as the high-value core for this group and the rest as a library to draw on by need.
https://r4ds.hadley.nz/

1. Applied R for Epidemiology (start here)

- **The Epidemiologist R Handbook** — a free, task-centered R reference written by applied epidemiologists, covering data management, visualization, and common epi tasks; the single best bridge for an epi team moving to R
- **Applied Epi** — the nonprofit behind the handbook; also offers live, instructor-led R training for epidemiologists
- **CRAN Task View: Epidemiology** — the curated, maintained list of R packages specifically for epidemiology (note: focused on infectious-disease and environmental epi, not generic statistics)
- **R4Epis** and **RECON (R Epidemics Consortium)** — community projects, linked from the handbook, with field-oriented templates and tools

> **TIP**
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> Point newcomers at the handbook’s table of contents and search box first — it is structured as "I need to do X," which matches how an analyst actually arrives at a question.

DEEP DIVE

The handbook is free and continuously updated, with an offline version for low-connectivity settings; it is the resource most likely to pay off immediately for this team. The CRAN Task View is the right place to discover a package for a specific epi task, using the same package-vetting habits from Lesson 19.
https://epirhandbook.com/

DEEP DIVE

Applied Epi's live training and the CRAN Epidemiology Task View are the two companion links worth bookmarking alongside the handbook. The Task View is maintained by working epidemiologists and curated by hand, so it is a trustworthy first stop.
https://appliedepi.org/ — and — https://cran.r-project.org/view=Epidemiology

2. R Packages for Epi & Biostatistics

- **Measures & two-by-two tables** — `epitools`, `epiR`, and `Epi` compute risk/odds ratios, rate ratios, attributable risk, and standardized rates with confidence intervals
- **Model results as tidy tables** — `broom` (from Day 4) tidies model objects; `gtsummary` produces publication-ready summary and regression tables; `parameters` (easystats) gives clean model summaries
- **Survival analysis** — the `survival` package is the standard for time-to-event models; `survminer` draws Kaplan–Meier curves
- **Modelling ecosystem** — `tidymodels` (from Day 4) unifies the broader modelling and machine-learning packages

WARNING

`epitools` / `epiR` make it easy to compute a rate ratio, but they do not decide whether a rate ratio is the right measure for your design. Keep that judgment with the analyst and statistician.

DEEP DIVE

Every R package has a stable page on CRAN at `cran.r-project.org/package=NAME` (for example `cran.r-project.org/package=epitools`), which links to its manual and vignettes — use that as the canonical, always-valid reference. `gtsummary` is especially worth a look for this team: it turns a fitted model into a formatted table suitable for a report or manuscript in a couple of lines, and it pairs naturally with the Day 5 reporting workflow.
https://cran.r-project.org/package=gtsummary

DEEP DIVE

For two-by-two and rate calculations without writing R, **OpenEpi** is a long-standing free web tool that many epidemiologists already know; it is handy for quick checks and teaching. The CRAN Task Views for **ClinicalTrials** and **MachineLearning** complement the Epidemiology view when the work moves toward trials or prediction.
https://www.openepi.com/

3. Foundational & Refresher Material

- **CDC, "Principles of Epidemiology in Public Health Practice" (Self-Study Course SS1978)** — the classic free CDC introduction to applied epidemiology and biostatistics; archived in January 2025 but still available through the CDC Stacks repository
- **"Gordis Epidemiology"** (Celentano & Szklo) — the standard introductory epidemiology textbook
- **"Modern Epidemiology"** (Rothman, Greenland & Lash) — the advanced reference for study design and causal inference
- **"Essentials of Epidemiology in Public Health"** (Aschengrau & Seage) — a widely used applied text
- **Statistics-in-R courses** — Imperial College's "Statistical Analysis with R for Public Health" (Coursera) and Johns Hopkins' epidemiology and data-science offerings (Coursera) teach the statistics and the R together

NOTE

This team already practices epidemiology, so treat this section as references and onboarding material for newer staff rather than as required reading.

DEEP DIVE

The CDC SS1978 course is the most directly relevant free refresher because it is written for federal, state, and local health professionals — the same setting as this team and their partners. Because CDC archived its original page in early 2025, link to the CDC Stacks copy rather than the old live URL, which no longer resolves.
https://stacks.cdc.gov/view/cdc/6914

4. Choosing & Interpreting Models (the IT/stats line)

- The course teaches how to **fit and extract** a model in R; deciding **which** model and reading **what it means** is a statistical and domain judgment
- Practical guidance for the team:
  - Bring a statistician in at **design** time, not just at analysis time — the hardest choices (confounding, study design, which measure) are made before any code is written
  - Be explicit about the question a model answers, and about its assumptions, before interpreting coefficients
  - Document modelling decisions in the report itself (Day 5), so reviewers and partners can see them
- The R skills from this week make that collaboration smoother: tidy output and reproducible reports give a statistician something clean to review

IMPORTANT

The goal of keeping Day 4 mechanical was not to avoid statistics but to put it where it belongs — with the people qualified to judge it. These resources support that collaboration; they are not a substitute for it.

DEEP DIVE

Tidy model output (`broom`), formatted tables (`gtsummary`), and a reproducible Quarto report together make a model auditable: a statistician can see the specification, the estimates, the confidence intervals, and the data handling in one document. That auditability is itself a contribution the R workflow makes to sound statistical practice.
https://broom.tidymodels.org/

5. Staying Current

- **CRAN Task Views** — re-check the Epidemiology, ClinicalTrials, and MachineLearning views periodically; they are updated as the ecosystem moves
- **The Epidemiologist R Handbook** — actively maintained, so revisit it for new chapters and updated code
- **Applied Epi and RECON communities** — for training, templates, and contact with other applied epidemiologists using R
- **The tidymodels and tidyverse sites** — for the modelling and data-wrangling tools the course is built on

> **TIP**
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> Pair this with `renv` (Lesson 19): when you adopt a new package from a Task View, snapshot it so your reports stay reproducible as the ecosystem changes around you.

DEEP DIVE

The ecosystem moves quickly, so the durable skill is knowing where to look — Task Views to discover packages, the handbook for applied recipes, and CRAN package pages to vet what you find. The vetting habits from Lesson 19 apply directly to everything in this document.
https://cran.r-project.org/web/views/

Summary

- **Start with applied R for epidemiology** — the Epidemiologist R Handbook, Applied Epi training, and the CRAN Epidemiology Task View are the highest-value resources for this team
- **Reach for vetted packages** — `epitools` / `epiR` / `Epi` for measures, `broom` / `gtsummary` for tidy and formatted results, `survival` for time-to-event, `tidymodels` for the broader ecosystem
- **Use the foundational material as references** — the CDC SS1978 course (via CDC Stacks) and standard texts, mainly for newer staff
- **Keep the IT/stats line** — fit and extract in R, but involve a statistician for design, method choice, and interpretation
- **Stay current** through the Task Views, the handbook, and the applied-epi community, and pin new packages with `renv`
- All package references resolve at `cran.r-project.org/package=NAME`, the stable canonical page for any CRAN package